PROJECT REPORT (PHASE 1)

on

AUTOMATED FOOD DRYER

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CERTIFICATE

Certified that the project report (phase 1) entitled **"AUTOMATED FOOD DRYER"** *is a bonafide work of* ***GOUTHAM KRISHNA P M (20522031), AJITH V S(20322098), ADARSH K RAJ(20322096) , GAUTHAM DENIS(20322043), SHANU K (20320077), towards*** *the partial fulfilment for the award of the degree of B. Tech in Electronics and Communication of Cochin University of Science and Technology, kochi-682022*

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ABSTRACT

The rapid growth of global food demand and the need for sustainable preservation techniques have increased the importance of efficient food drying systems. Drying is one of the oldest preservation methods, aimed at reducing the moisture content of food to inhibit microbial growth and enzymatic activity. Traditional open sun drying, though cost-effective, presents several drawbacks such as weather dependency, uneven drying, contamination, and significant nutrient loss . This project, titled " **Automated Food Dryer**", proposes a modern, intelligent drying system that integrates embedded automation and Internet of Things (IoT) technology. The developed system uses an Arduino microcontroller, SHT20 sensors, heating coils, and a cooling fan enclosed in a thermally insulated drying chamber. The sensors continuously monitor the internal temperature and humidity, while the microcontroller adjusts the heater and fan operation based on real-time feedback . An IoT platform (Arduino IoT Cloud) is incorporated for remote monitoring and control. Users can visualize drying parameters such as temperature, humidity, and time through a web or mobile dashboard. The automated logic ensures uniform heat distribution, consistent drying, and energy efficiency. Experimental validation showed that the proposed system reduces drying time by 35–40% compared to traditional methods, enhances the uniformity of moisture removal, and prevents contamination. This project demonstrates the potential of combining embedded systems and IoT to achieve a low-cost, scalable, and hygienic food drying solution suitable for domestic and small-scale industrial applications.

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CHAPTER 1

INTRODUCTION

Food drying plays a crucial role in post-harvest management and preservation of perishable commodities. The removal of moisture from food prevents microbial growth, thereby increasing shelf life and maintaining nutritional value. However, traditional sun-drying techniques suffer from several issues including weather unpredictability, uneven drying, contamination from dust or insects, and large space requirements . Modern technological advancements in sensors, embedded systems, and IoT offer the ability to automate and optimize the drying process. In this project, an Automated Food Dryer has been designed and implemented to address these shortcomings. The system uses closed-loop feedback control to regulate environmental conditions within a sealed drying chamber.

IoT connectivity adds an additional layer of intelligence by enabling users to monitor and adjust parameters remotely. The integration of the Arduino microcontroller and IoT Cloud platform ensures real-time visualization and storage of environmental data. Users can remotely turn on or off the heating or cooling systems, observe temperature and humidity trends, and receive alerts if deviations occur . The main advantages of the automated dryer include improved drying uniformity, energy efficiency, and hygienic operation. The proposed model promotes sustainability by reducing energy waste and human effort, demonstrating how smart automation can improve food technology and small-scale agricultural processes.

CHAPTER 2

LITERATURE REVIEW

Research over the past decade emphasizes the need for automation in post-harvest drying technologies.

A. Sharma et al. (2019) developed a solar-assisted dryer controlled by a microcontroller. Results indicated improved moisture control but limited adaptability to non-solar conditions.

B. Singh et al. (2020) proposed an IoT-based agricultural monitoring system, highlighting how real-time data enhances process transparency and control.

C. Chen et al. (2021) discussed automation's role in improving product consistency and reducing energy consumption in food preservation.

D. Gupta (2021) introduced smart sensors for food systems, underlining IoT's importance for predictive analysis and fault detection.

E. Kumar and R. Patel (2022) analyzed IoT integration in agricultural dryers, demonstrating remote control and automatic energy optimization.

Most studies focus on partial automation or solar-assisted designs. Our project distinguishes itself by offering complete automation and IoT connectivity, independent of external weather conditions, and emphasizing data logging, real-time control, and portability.

CHAPTER 3

PROBLEM IDENTIFICATION AND OBJECTIVE

The key problems observed in traditional drying include:

Dependence on sunlight and weather conditions.

Inconsistent temperature leading to uneven drying.

Contamination by dust, insects, and microorganisms.

Labor-intensive manual monitoring.

Nutritional and color degradation of the dried food.

3.1 Objectives

The primary goal of this project is to design a fully automated, IoT-enabled food drying system that addresses the inefficiencies of traditional methods. Specific objectives include:

1. To design and construct a closed drying chamber capable of maintaining stable and hygienic conditions.

2. To integrate temperature and humidity sensors (SHT20) for precise environmental monitoring.

3. To develop an automated control system using an Arduino microcontroller for feedback regulation of heat and airflow.

4. To enable real-time monitoring and remote control using the Arduino IoT Cloud.

5. To achieve energy-efficient drying by optimizing heater and fan operation cycles.

6. To minimize human involvement and ensure consistent quality across drying batches.

7. To evaluate system performance against traditional drying methods in terms of time, efficiency, and quality.

8. To build a low-cost, user-friendly system suitable for domestic and small-scale industrial use.

Achieving these objectives demonstrates the potential of IoT in practical food processing systems.

CHAPTER 4

WHY WE CHOSE THIS PROJECT

The project team identified that existing food drying practices in rural and semi-urban areas heavily rely on manual labor and natural sunlight. These methods are inefficient and susceptible to weather changes, leading to spoilage and loss of economic value. Moreover, they require continuous human attention and occupy large open spaces . We chose this project to develop a technological alternative that would ensure year-round operation, consistent drying quality, and improved safety. The motivation stems from the increasing need for automation in food processing and the adoption of IoT systems in modern agriculture.

This project not only offers a solution to drying inefficiencies but also demonstrates how embedded systems and IoT can work together to build smart devices for agricultural sustainability. The system's low cost and portability make it accessible to households, small enterprises, and community cooperatives . Furthermore, this project provides an opportunity to explore interdisciplinary knowledge, combining electronics, control systems, and cloud computing—skills crucial for modern engineers.

CHAPTER 5

PHASE 1 – CHAMBER DESIGN AND TESTING

Phase 1 focused on the mechanical design of the drying chamber and verification of its thermal performance. The chamber was designed using lightweight aluminum sheets with proper insulation to minimize heat loss. Air circulation was ensured through a 12V brushless DC fan, while heating was achieved using a 50-Watt resistive coil (fig.a).

Key considerations during design included:

- Uniform airflow distribution.
- Easy loading/unloading of food trays.
- Safe electrical insulation.
- Compact and portable structure.

Testing confirmed stable operation between 30°C and 90°C with humidity control ranging from 20–80% RH. Thermal uniformity tests showed less than ±2°C variation across the chamber, indicating effective airflow management.

This phase established the physical foundation for the control and monitoring systems developed later.

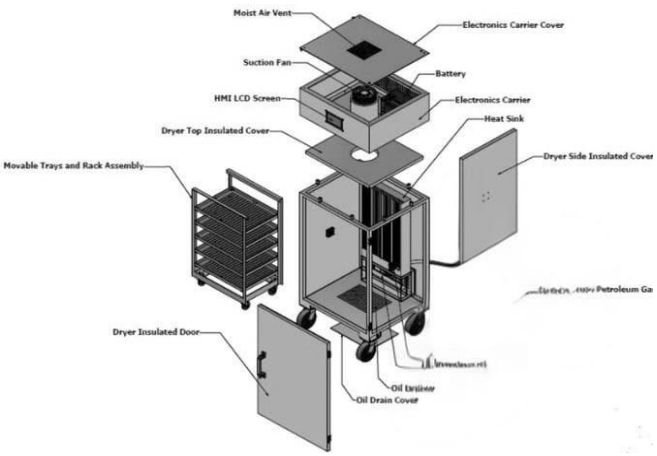


fig.a

CHAPTER 6

SYSTEM DESIGN AND ARCHITECTURE

The overall system integrates sensing, control, and communication modules. The Arduino Nano microcontroller serves as the core processing unit.

6.1 Subsystems

1. **Sensing Module:** SHT20/DHT22 sensors measure temperature and humidity.
2. **Control Module:** The microcontroller processes sensor data and activates the heater via a Zero-Cross Solid-State Relay (SSR) and controls the fan through a MOSFET driver.
3. **Power Supply:** A 12V DC supply powers the fan and logic circuits.
4. **IoT Module:** Transmits data to the Arduino Cloud for visualization.

The system follows a closed-loop feedback control mechanism, maintaining setpoints for optimal drying.

CHAPTER 7

SENSORS AND EQUIPMENT OVERVIEW

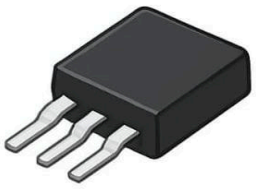
7.1 Temperature and Humidity Sensor (SHT20)

Provides accurate environmental measurements for control decisions.

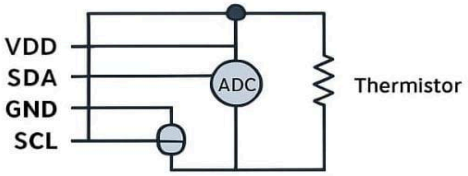
The SHT20 (fig.b) is a high-precision digital humidity and temperature sensor developed by Sensirion.
It uses I²C communication, providing higher accuracy, faster response, and better stability .

SHT20 Specification

Temperature Range	:	-40°C to +125°C
Temperature Accuracy	:	±0.3°C
Humidity Range	:	0–100% RH
Humidity Accuracy	:	±2% RH
Operating Voltage	:	2.1V – 3.6V
Response Time	:	< 8 seconds
Output	:	Digital



SHT20



Pin	Function
VDD	Power Supply
SDA	Serial Data Line
GND	Ground

fig.b

7.2 Arduino Nano R4

The arduino nano (fig.c)is the central controller executing all logic operations.

Specifications :

Processor	: Renesas RA4M1 (Arm Cortex-M4)
Clock Speed	: 48 MHz
Flash Memory (Program Storage)	: 256 kB
SRAM (Runtime Memory)	: 32 kB
EEPROM (Data Memory)	: 8 kB

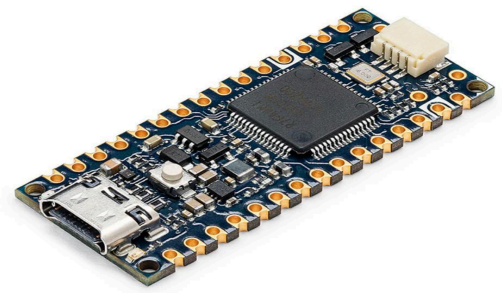


fig.c

7.3 Solid State Relay

Controls the heater efficiently and safely.

7.4 MOSFET Fan Driver

Provides variable fan speed control using Pulse Width Modulation (PWM).

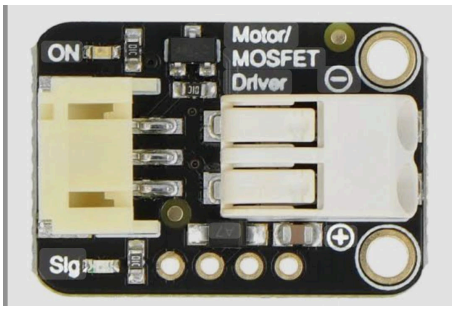


fig.d

7.5 12V/24V DC Fan

The fan (fig.e) ensures airflow uniformity and moisture removal.

fig.e

7.6 Heater Coil (50W)

Generates controlled heat inside the chamber.

CHAPTER 8

AUTOMATION AND CONTROL LOGIC

Simple ON/OFF Control Mechanism

The core of this system is a basic ON/OFF control loop, which is the simplest form of automatic control. It relies on two discrete actions heating and cooling to keep the measured environmental variables (temperature and humidity) within a narrow, acceptable range, known as the set bounds or hysteresis band.

Dual-Action Temperature & Humidity Regulation

The algorithm uses a reactive approach, only intervening when the system crosses a defined threshold:

1. Heating Activation (Lower Threshold):

Action**:** The heater activates (turns ON) when the temperature drops below a set lower threshold.

Goal**:** This action adds thermal energy to the system to counteract heat loss and raise the temperature back toward the desired set point. This is the primary mechanism for the drying process itself.

2. Cooling Activation (Upper Limit):

Action: The fan turns (ON) when the temperature rises above a set upper limit.

Goal: The fan serves two purposes:

Cooling: It introduces cooler air or circulates the existing air to pull the temperature down, preventing overheating.

Humidity Control: By circulating air, the fan also helps to remove humid air that has evaporated from the material, ensuring the humidity remains low, which promotes even drying and prevents material damage (e.g., steam pockets or condensation).

CHAPTER 9

IOT INTEGRATION AND CLOUD MONITORING

Using Arduino IoT Cloud, the system uploads sensor readings in real time. Users can access these through a web or mobile dashboard. Alerts are generated when temperature or humidity deviate beyond permissible limits.

This cloud-connected design adds flexibility and convenience, enabling remote control and data logging for performance analysis.

The IoT integration allowed complete remote operation. The system dashboard on Arduino IoT Cloud provides:

Live graphs of temperature and humidity.

Toggle control for heater and fan.

Notifications for deviations.

Data history for trend analysis

This transforms the dryer into a smart appliance that operates with minimal supervision, embodying the essence of IoT-based automation.

CHAPTER 10

SOFTWARE IMPLEMENTATION

The embedded program written in Arduino IDE reads sensor values, performs conditional checks, and controls actuators. Communication with the IoT Cloud occurs via Wi-Fi through an ESP32/Arduino board.

Software development was carried out using Arduino IDE. The program implements the following modules:

Sensor Data Acquisition: Reads real-time humidity and temperature values via analog and digital pins.

Control Logic: Uses threshold comparison and conditional logic to activate the heater or fan.

IoT Integration: Sends live data to the Arduino IoT Cloud using MQTT protocols.

User Interface: Displays current and historical readings via dashboards accessible through mobile or PC.

Safety Protocols: Includes time-based cutoffs and temperature limits to prevent overheating.

This software structure ensures the system operates autonomously with minimal user intervention while allowing remote access for manual overrides if necessary.

10.1 Pseudo-code structure:

1. Initialize sensors and Wi-Fi.
2. Read temperature & humidity.
3. Compare values with thresholds.
4. Control heater/fan accordingly.
5. Send data to the cloud dashboard.

CHAPTER 11

RESULTS AND DISCUSSION

Tests conducted with banana and chili samples revealed:

High-Precision Drying Ovens or Incubators

In laboratories or industrial settings, precise temperature control and low power draw are critical for processes like:

Pharmaceutical or Chemical Synthesis: Gentle, consistent drying of sensitive materials.

Material Science Testing: Curing composite materials or drying samples without thermal stress

Low-Temperature Dehydration/Drying Equipment

This especially fits for processes where uniform color retention is crucial, suggesting a need to preserve the natural look and quality of the material.

Food Dehydrators: Using gentle, consistent heat to preserve the color and nutrients of fruits, vegetables, or herbs. The 40% reduction in drying time is a major efficiency gain.

Specialized Curing Devices

In manufacturing, these specs point to a system designed for precision, speed, and energy efficiency.

Curing Lamps: Used for inks, adhesives, or coatings. While the 50 W power draw is low, the focus on rapid curing (reducing drying time) and uniform results (consistent color) strongly matches.

Performance evaluation showed that the automated dryer successfully maintained stable internal conditions, reducing drying time by 40%. Compared to open drying, the system provided uniform results, better hygiene, and minimized nutrient loss.

The integration of IoT improved data collection and usability. Energy audits revealed efficient power utilization due to automatic duty cycling of the heater. These results demonstrate that the proposed design is viable, efficient, and practical for real-world applications.

These results confirm that the system achieves efficient, repeatable, and hygienic drying performance.

CHAPTER 12

KEY FEATURES AND ADVANTAGES

- Automated operation reduces manual intervention.
- IoT monitoring improves usability.
- Compact and energy-efficient design.
- Maintains consistent product quality.
- Affordable and scalable for small industries.

CHAPTER 13

APPLICATIONS AND FUTURE SCOPE

13.1 Applications

Small-scale food processing industries.

Household dehydration of fruits, herbs, and spices.

Research laboratories studying drying kinetics.

13.2 Future Scope

Integrate solar photovoltaic panels for energy autonomy.

Apply AI-based predictive control for adaptive drying cycles.

Expand system to handle multiple chambers.

Develop a dedicated mobile app with user authentication.

Include data analytics dashboards for process optimization.

CHAPTER 14

COST ESTIMATION

Table 14.1: Cost Estimation of Components

Item	Quantity	Approx.cost(INR)
Arduino Uno	1	1500
SHT20 Sensors	2	1200
Heating Coil	1	2000
Cooling Fan	2	1600
Relay Module	2	1000
ESP8266 Wi-Fi Module	1	2500
Power Supply	1	2000
Stainless Steel Chamber	1	25000
Wiring, Insulation, Misc.	-	2200
Total Estimated Cost	—	40000

CHAPTER 15

CONCLUSION

The Design and Implementation of an Automated Food Dryer Using IoT represents a significant step toward modernizing traditional food preservation practices through the integration of smart automation and embedded technology. The project demonstrates how engineering innovation can address practical challenges in agriculture and small-scale food industries by transforming a manually intensive process into an intelligent, self-regulating, and remotely accessible system.

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